

PROJECT 1 STATIC ROUTING

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TASK 1: CREATING STATIC ROUTES IN THE ROUTER

ROUTER 1

R1 IP: 10.0.0.1 – eth0/0

STATIC ROUTE R1:

15.0.0.0/8 20.0.0.2

21.0.0.0/8 20.0.0.2

30.0.0.0/8 20.0.0.2

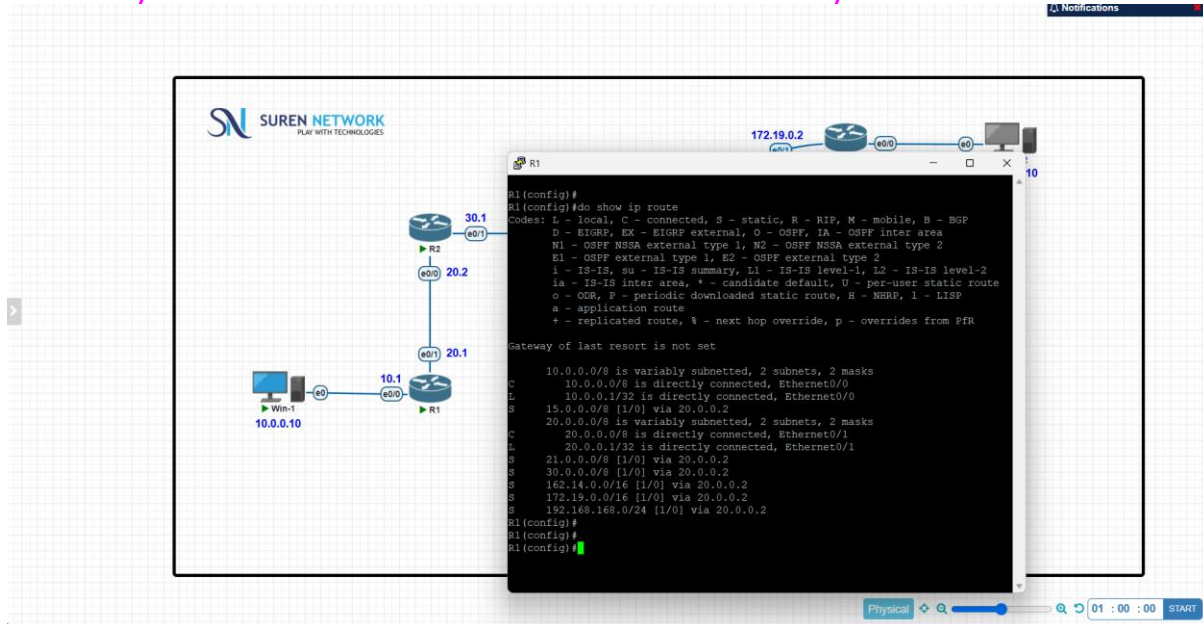
R1 IP: 20.0.0.1 – eth0/1

STATIC ROUTE R1:

162.14.0.0/16 20.0.0.2

172.19.0.0/16 20.0.0.2

192.168.168.0/24 20.0.0.2



ROUTER 2

R2 IP: 20.0.0.2 – eth0/0

STATIC ROUTE R2:

10.0.0.0/8 20.0.0.1

15.0.0.0/8 30.0.0.2

21.0.0.0/8 30.0.0.2

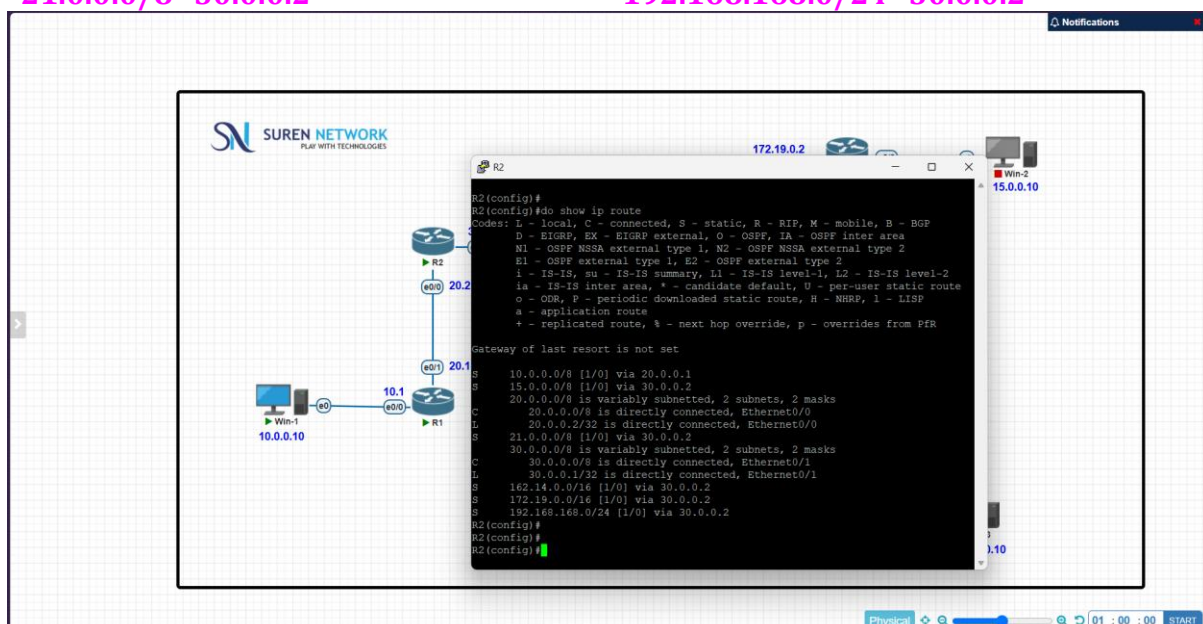
R2 IP: 30.0.0.1 – eth0/1

STATIC ROUTE R2:

162.14.0.0/16 30.0.0.2

172.19.0.0/16 30.0.0.2

192.168.168.0/24 30.0.0.2



ROUTER 3

R3 IP: 30.0.0.2 - eth0/0

R3 IP: 172.19.0.1 - eth0/1

R3 IP: 21.0.0.1 - eth0/2

STATIC ROUTE R3:

10.0.0.0/8 30.0.0.1

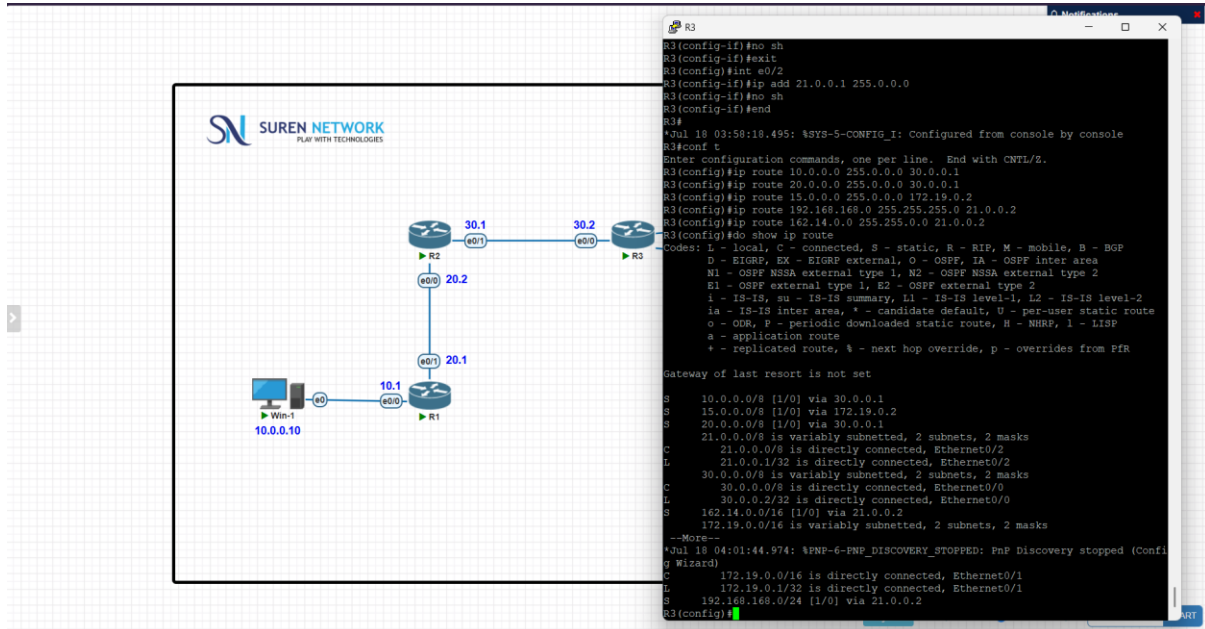
15.0.0.0/8 172.19.0.2

20.0.0.0/8 30.0.0.2

STATIC ROUTE R3:

162.14.0.0/16 21.0.0.2

192.168.168.0/24 21.0.0.2



ROUTER 4

R4 IP: 15.0.0.1 - eth0/0

R4 IP: 172.19.0.2 - eth0/1

STATIC ROUTE R4:

10.0.0.0/8 172.19.0.1

20.0.0.0/8 172.19.0.1

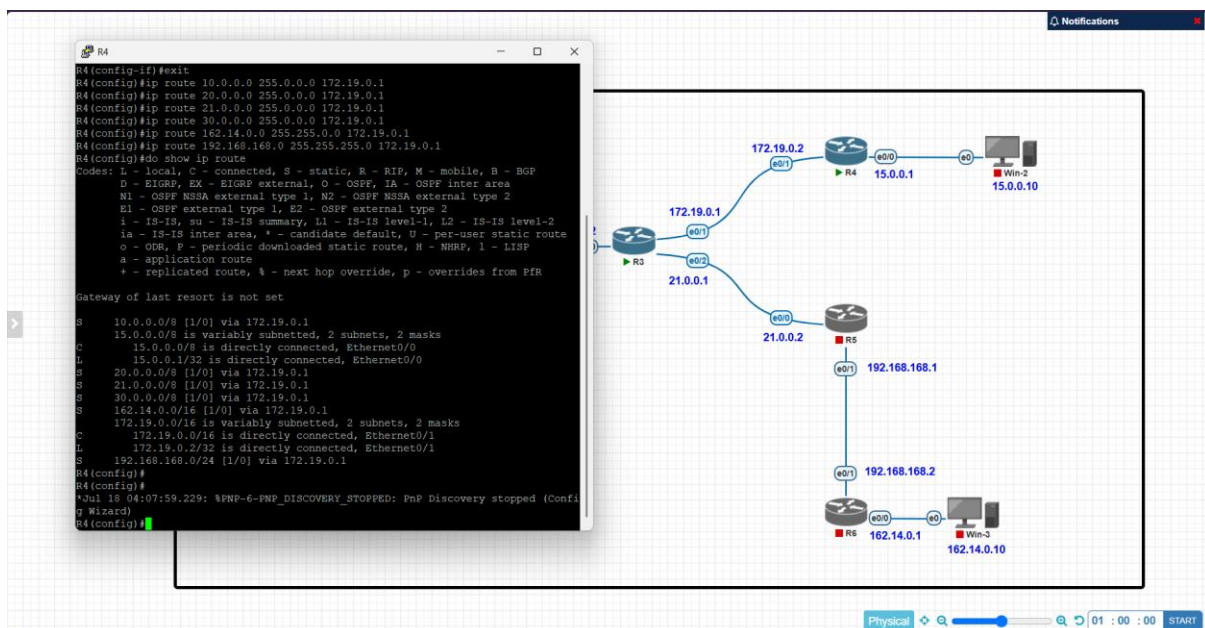
21.0.0.0/8 172.19.0.1

STATIC ROUTE R4:

30.0.0.0/8 172.19.0.1

162.14.0.0/16 172.19.0.1

192.168.168.0/24 172.19.0.1



ROUTER 5

R5 IP: 21.0.0.2 – eth0/0

STATIC ROUTE R5:

10.0.0.0/8 21.0.0.1

15.0.0.0/8 21.0.0.1

20.0.0.0/8 21.0.0.1

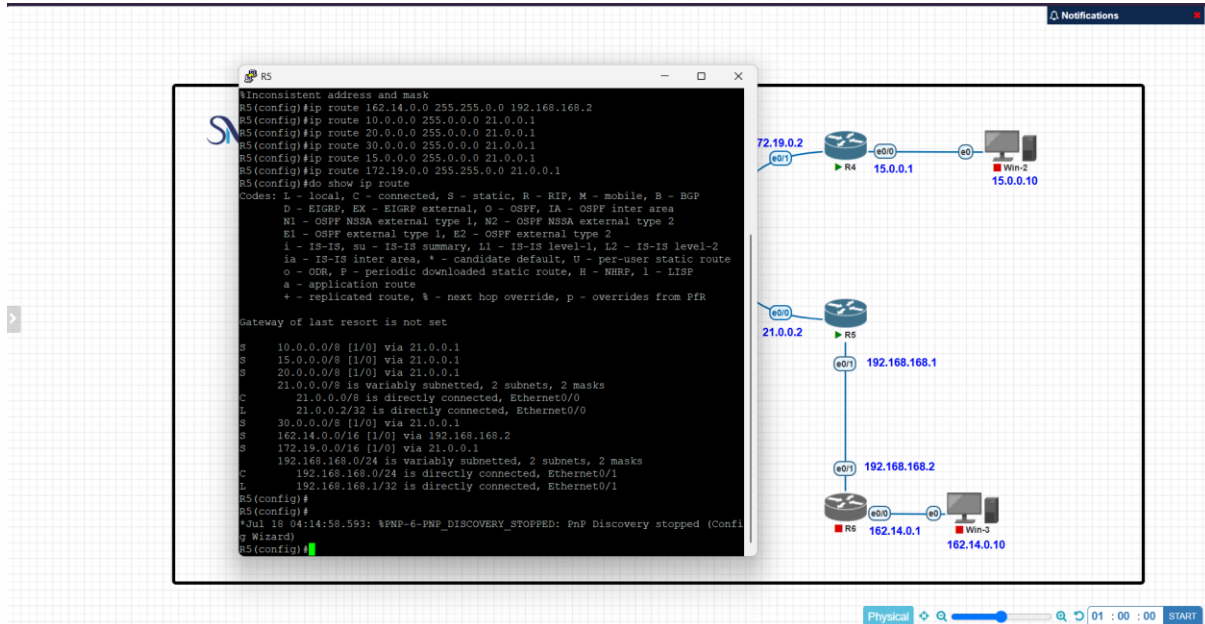
R5 IP: 192.168.168.1 – eth0/1

STATIC ROUTE R5:

30.0.0.0/8 21.0.0.1

162.14.0.0/16 192.168.168.2

172.19.0.0/16 21.0.0.1



ROUTER 6

R6 IP: 162.14.0.1 – eth0/0

STATIC ROUTE R6:

10.0.0.0/8 192.168.168.1

15.0.0.0/8 192.168.168.1

20.0.0.0/8 192.168.168.1

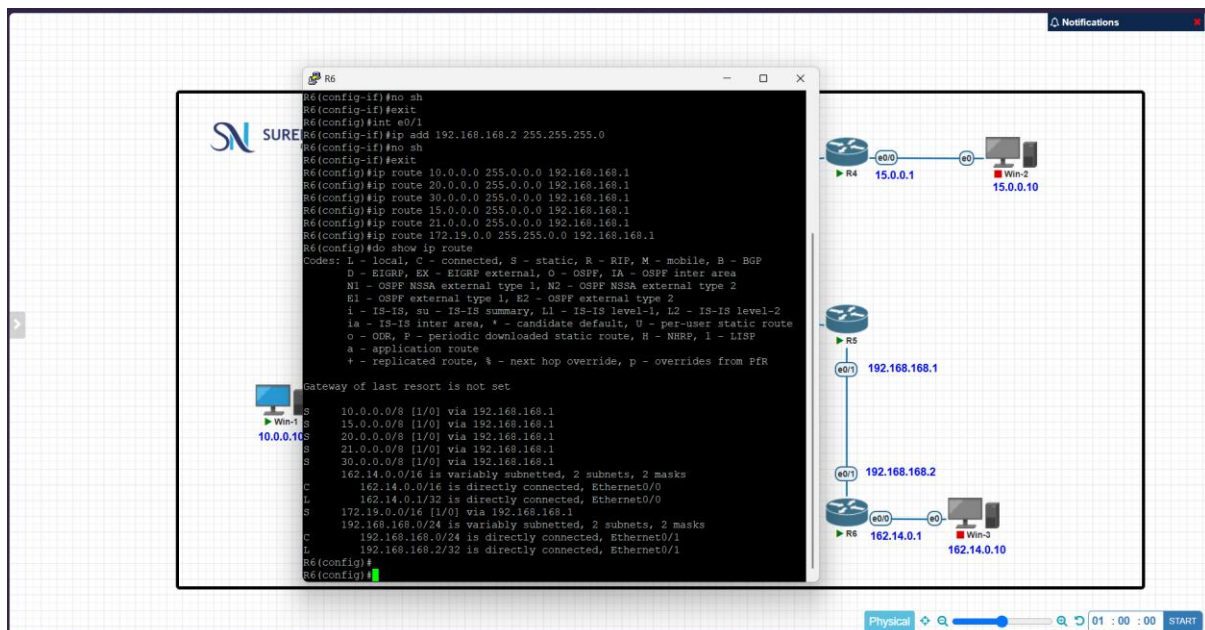
R6 IP: 192.168.168.2 – eth0/1

STATIC ROUTE R6:

21.0.0.0/8 192.168.168.1

30.0.0.0/8 192.168.168.1

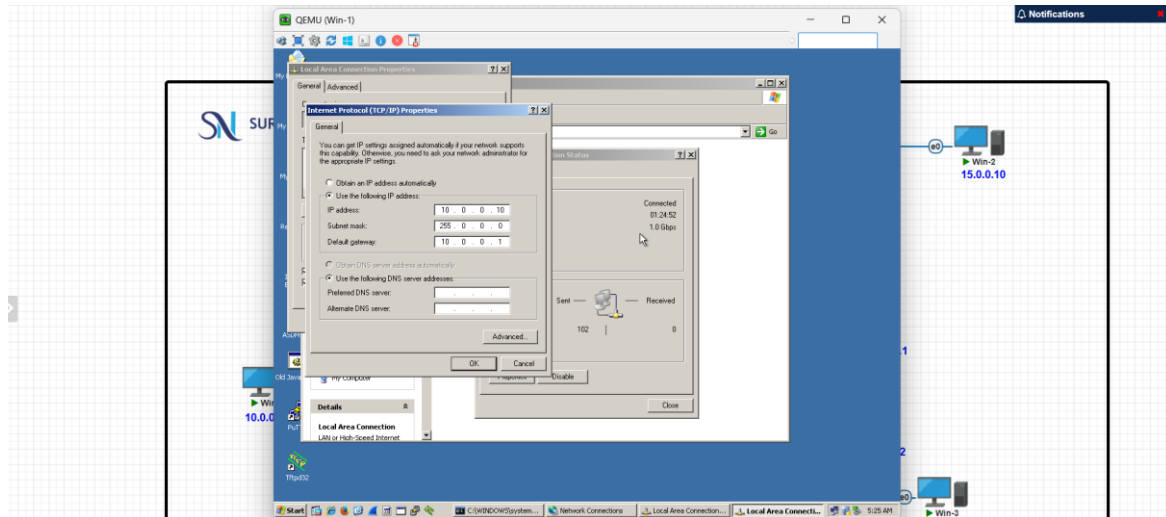
172.19.0.0/16 192.168.168.1



TASK 2: ASSIGNING IP ADDRESS TO PCs

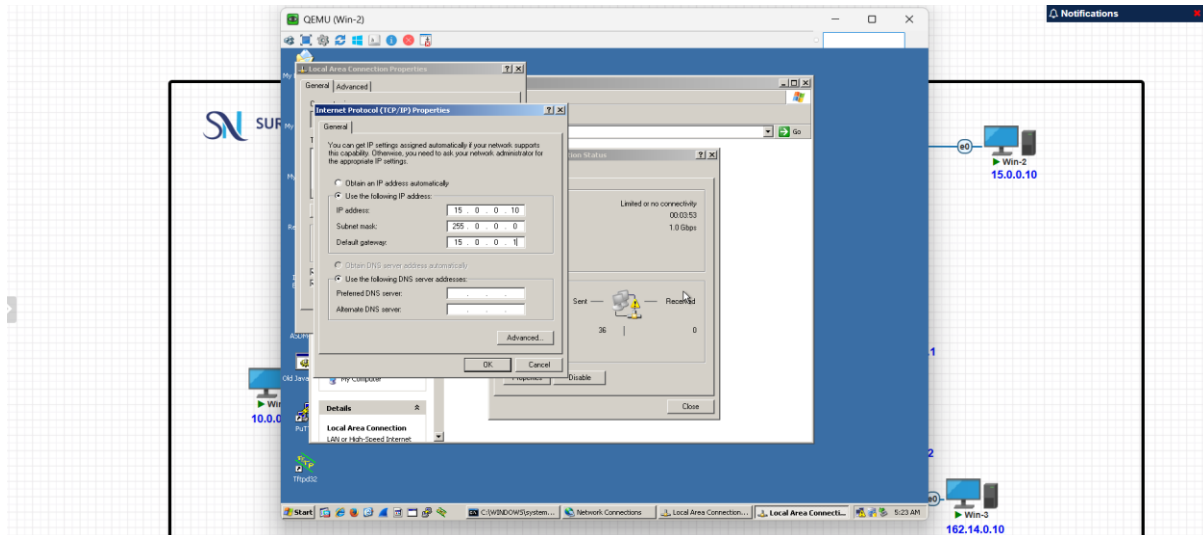
PC1 IP: 10.0.0.10

GATEWAY: 10.0.0.1



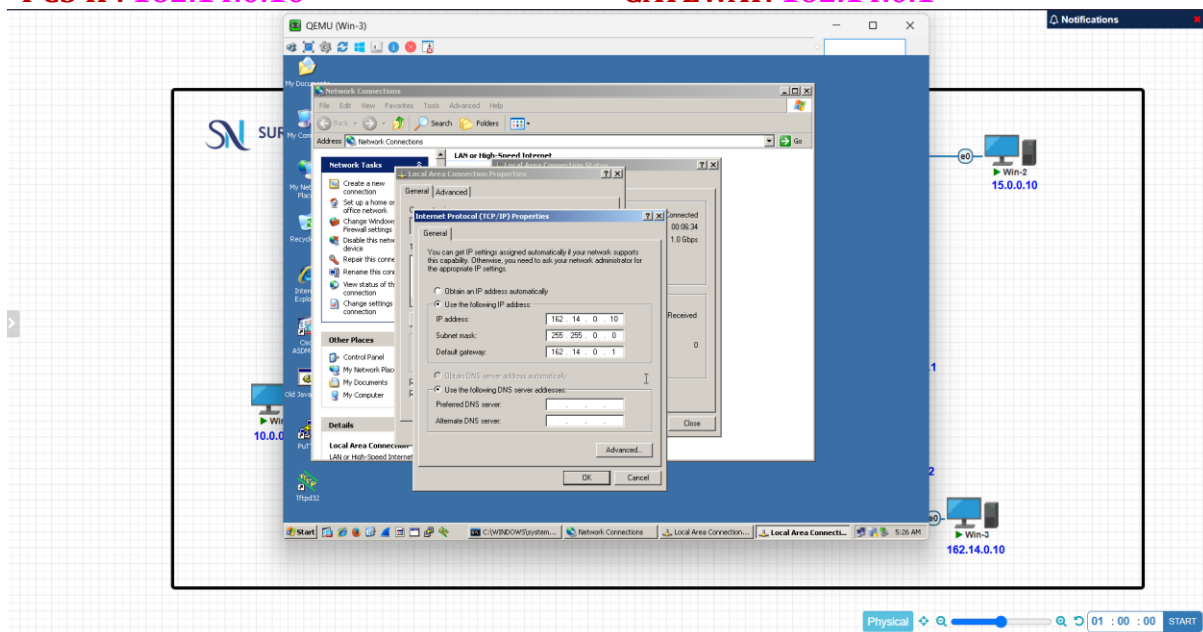
PC2 IP: 15.0.0.10

GATEWAY: 15.0.0.1



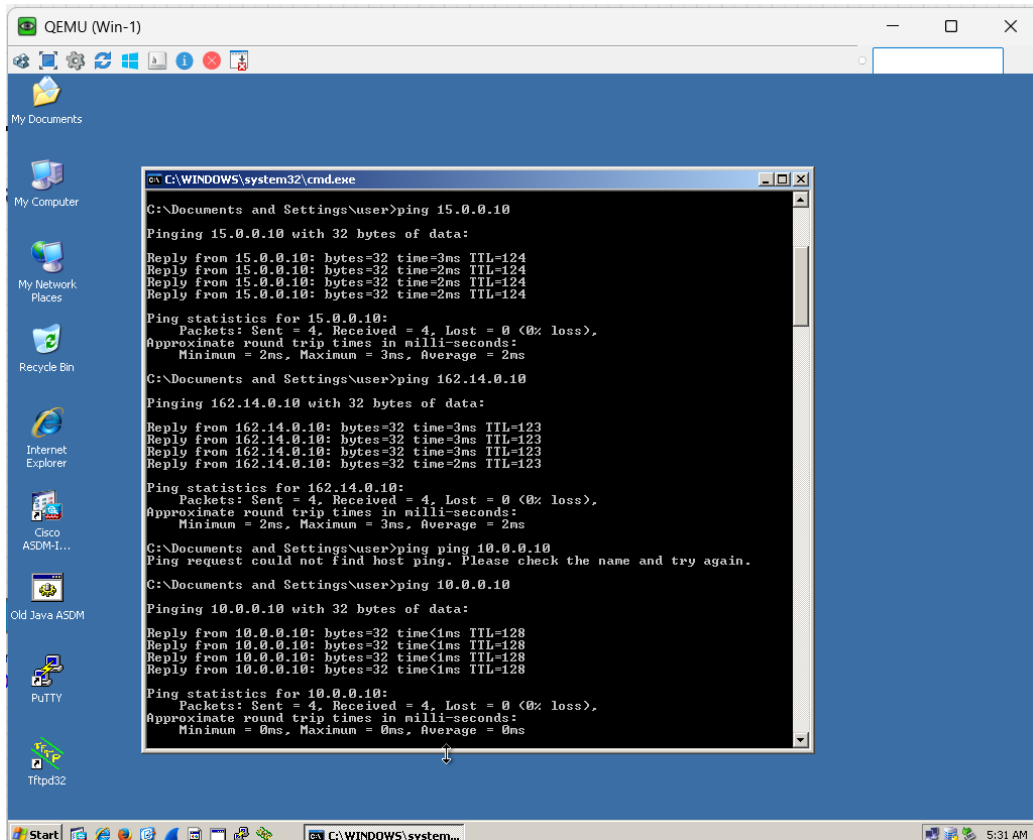
PC3 IP: 162.14.0.10

GATEWAY: 162.14.0.1



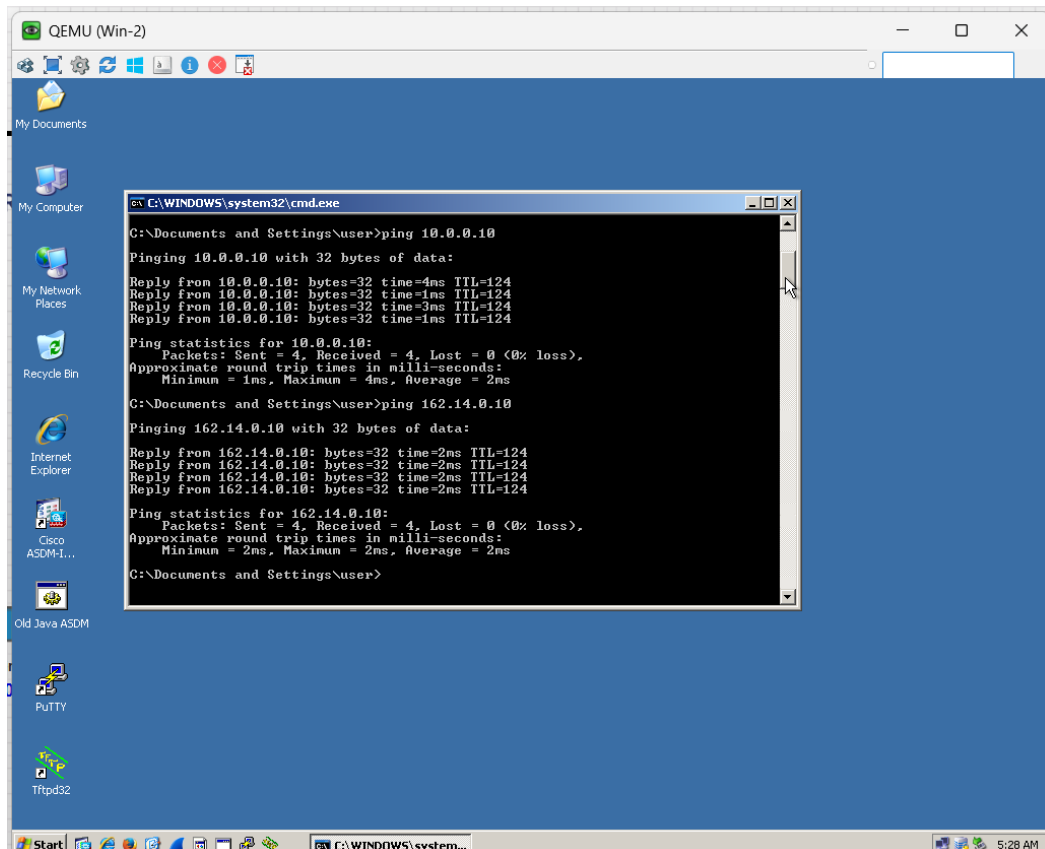
TASK 3: VERIFYING THE CONNECTION ESTABLISHMENTS

ping 10.0.0.10 (PC1) to 15.0.0.10 (PC2) & 10.0.0.10 (PC1) to 162.14.0.10 (PC3)



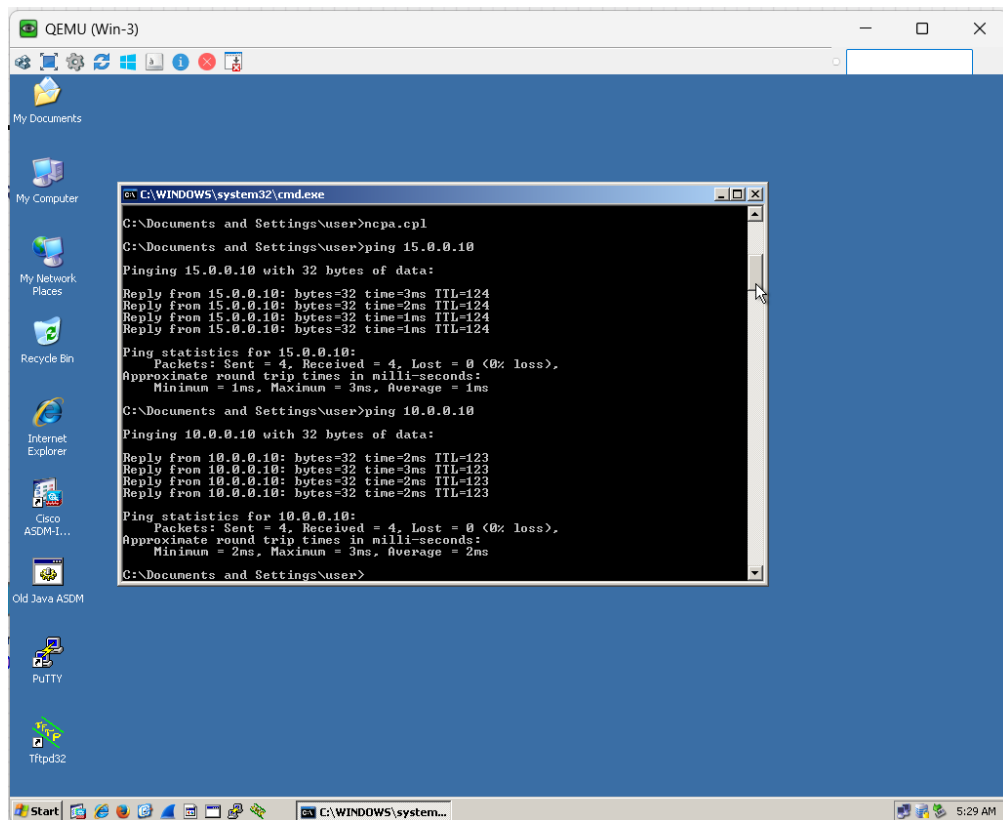
```
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\user>ping 15.0.0.10
Pinging 15.0.0.10 with 32 bytes of data:
Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms
C:\Documents and Settings\user>ping 162.14.0.10
Pinging 162.14.0.10 with 32 bytes of data:
Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123
Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms
C:\Documents and Settings\user>ping ping 10.0.0.10
Ping request could not find host ping. Please check the name and try again.
C:\Documents and Settings\user>ping 10.0.0.10
Pinging 10.0.0.10 with 32 bytes of data:
Reply from 10.0.0.10: bytes=32 time<1ms TTL=128
Reply from 10.0.0.10: bytes=32 time<1ms TTL=128
Reply from 10.0.0.10: bytes=32 time<1ms TTL=128
Reply from 10.0.0.10: bytes=32 time<1ms TTL=128
Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

ping 15.0.0.10 (PC2) to 10.0.0.10 (PC1) & 15.0.0.10 (PC2) to 162.14.0.10 (PC3)



```
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\user>ping 10.0.0.10
Pinging 10.0.0.10 with 32 bytes of data:
Reply from 10.0.0.10: bytes=32 time=4ms TTL=124
Reply from 10.0.0.10: bytes=32 time=1ms TTL=124
Reply from 10.0.0.10: bytes=32 time=3ms TTL=124
Reply from 10.0.0.10: bytes=32 time=1ms TTL=124
Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 4ms, Average = 2ms
C:\Documents and Settings\user>ping 162.14.0.10
Pinging 162.14.0.10 with 32 bytes of data:
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms
C:\Documents and Settings\user>
```

ping 162.14.0.10 (PC3) to 15.0.0.10 (PC2) & 162.14.0.10 (PC3) to 10.0.0.10 (PC1)

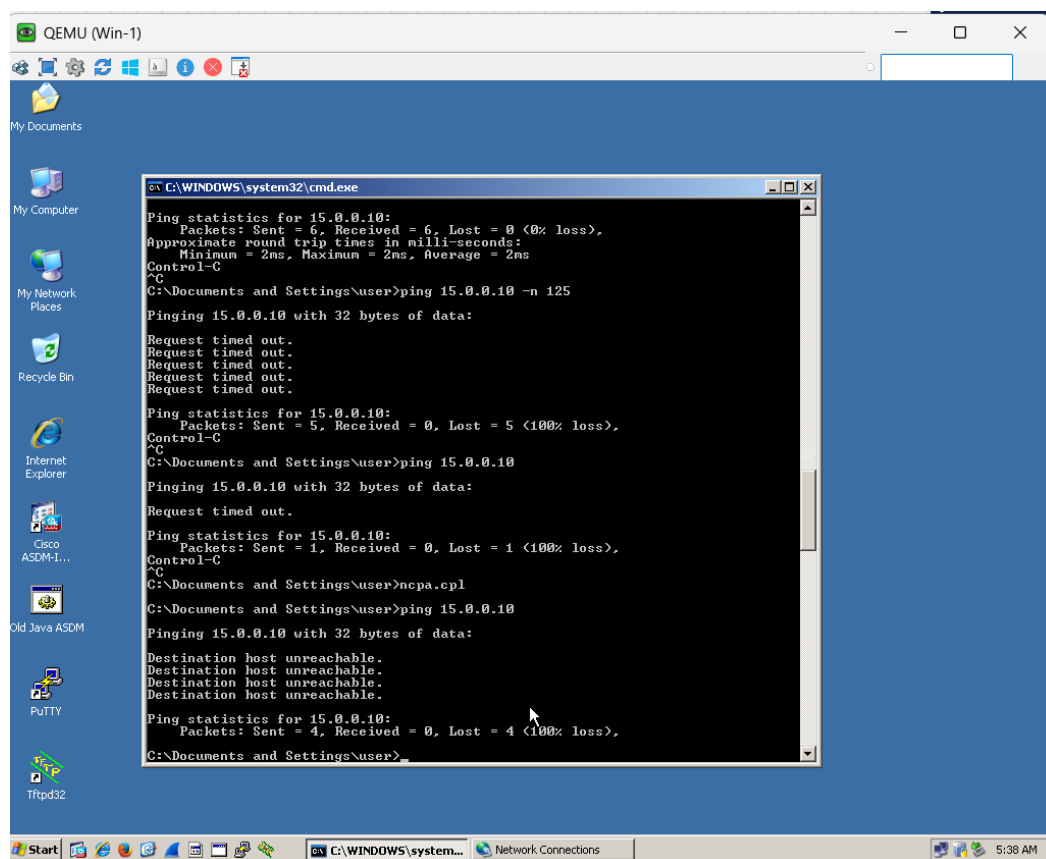


```
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\user>ncpa.cpl
C:\Documents and Settings\user>ping 15.0.0.10
Pinging 15.0.0.10 with 32 bytes of data:
Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=1ms TTL=124
Reply from 15.0.0.10: bytes=32 time=1ms TTL=124
Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 1ms
C:\Documents and Settings\user>ping 10.0.0.10
Pinging 10.0.0.10 with 32 bytes of data:
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=3ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms
C:\Documents and Settings\user>
```

TASK 4: IDENTIFYING THE ERRORS

ERROR 1: Request timed out. -> Data Packet sent but waiting for reply.

ERROR 2: Destination host unreachable. -> There is no route for the Destination IP in the source PC route table.



```
C:\WINDOWS\system32\cmd.exe
Ping statistics for 15.0.0.10:
    Packets: Sent = 6, Received = 6, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms
Control-C
^C
C:\Documents and Settings\user>ping 15.0.0.10 -n 125
Pinging 15.0.0.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 15.0.0.10:
    Packets: Sent = 5, Received = 0, Lost = 5 (100% loss),
Control-C
^C
C:\Documents and Settings\user>ping 15.0.0.10
Pinging 15.0.0.10 with 32 bytes of data:
Request timed out.
Ping statistics for 15.0.0.10:
    Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),
Control-C
^C
C:\Documents and Settings\user>ncpa.cpl
C:\Documents and Settings\user>ping 15.0.0.10
Pinging 15.0.0.10 with 32 bytes of data:
Destination host unreachable.
Destination host unreachable.
Destination host unreachable.
Destination host unreachable.
Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\Documents and Settings\user>
```